

EB-JEPA for EEG — Concepts and Intuitions

A pure-concept summary, no maths

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What is a representation?

An EEG recording is a sequence of voltages across 19 electrodes over time. These raw numbers are hard to reason about directly — they vary enormously across patients, devices, and recording conditions.

A **representation** is a compact summary of the recording: a vector of, say, 256 numbers that captures what *matters* about the signal and ignores what doesn't. A good representation is one where recordings that are clinically similar (both abnormal, or both showing the same type of event) end up close to each other in that 256-dimensional space, and clinically different recordings end up far apart.

The **encoder** is the function that maps a raw EEG window to its representation. Training an encoder is training it to produce these meaningful summaries.

Why self-supervised? The labelling bottleneck

To train a supervised model you need labelled examples: a neurologist must inspect each recording and write “normal” or “abnormal”. This is expensive, slow, and requires rare expertise. Most EEG data in the world is unlabelled.

Self-supervised learning (SSL) sidesteps this: it trains an encoder to produce good representations *without any labels*, using only the structure of the raw signal itself. The idea is to give the encoder a task that forces it to learn useful structure — not “is this abnormal?” but something like “do these two corrupted versions of the same signal look like the same thing?”

Once the encoder is trained on unlabelled data, a very simple classifier (a single linear layer, called a *probe*) is trained on a small labelled set. If the representations are good, the probe can separate normal from abnormal without ever retraining the encoder.

The cave allegory

In Plato's allegory of the cave, prisoners see only shadows projected on a wall. They have never seen the real objects casting the shadows, but they learn to recognise and predict the shadow patterns.

Our SSL encoder is in the same position. It sees raw EEG voltages — the shadows. It never sees the neurologist's verdict — the real object behind the signal. But by learning that *two different corrupted views of the same shadow are shadows of the same object*, it is forced to build an internal model of what is really there.

A linear probe then asks: “given your model of what is really there, is this patient’s brain behaving normally?”

What is invariance? What is collapse?

Invariance is the property we want: the encoder should output the same representation for two corrupted versions of the same signal. Forcing $z_1 \approx z_2$ (where z_1 and z_2 are the representations of the two views) is the training signal.

The problem: the trivially simplest way to satisfy $z_1 = z_2$ is to output the **same constant vector for everything**. This is called **collapse** — the encoder learns nothing, because all signals look the same.

Anti-collapse regularisation prevents this. VICReg does it by adding two extra penalties: one that forces the encoder to *spread* its output across dimensions (variance term), and one that forces different output dimensions to carry different information (covariance term). SIGReg does it by testing that the distribution of representations looks Gaussian (a non-constant distribution cannot be collapsed).

Can predictive architectures avoid collapse without regularisation? There is a common claim that in I-JEPA-style models — where the encoder predicts the representation of a *different, disjoint* part of the input — collapse is impossible: “if the encoder outputs a constant, the MSE to the target cannot be zero.” This is *not quite right*. If both the online and EMA encoders collapse to the same constant c , the predictor can simply learn to output c too, and the loss is zero.

What actually prevents collapse in practice is a combination of: (1) **BatchNorm** — normalisation layers make a true constant output numerically explosive; (2) **EMA momentum** — the target encoder lags behind the online encoder, creating a non-stationary signal that is hard to collapse onto; and (3) **input diversity** from the disjoint split, which gives useful gradient at the start of training. The temporal split *helps*, but is not the full story.

What does the score mean?

BACC (balanced accuracy) is the average of the true positive rate (how often abnormal recordings are correctly detected) and the true negative rate (how often normal recordings are correctly classified). It is fair even when the two classes are unequal in size. Random guessing gives 0.5; perfect classification gives 1.0.

AUROC (area under the ROC curve) measures how well the model *ranks* recordings by their abnormality score, at every possible threshold. 1.0 is perfect; 0.5 is random.

Per-window vs per-recording: a recording is typically several minutes long, cut into 10-second windows. Per-window scoring classifies each window independently; per-recording scoring averages across all windows of the same patient. Per-recording scores are systematically about 5 percentage points higher for the same encoder, because averaging reduces noise. Most of the published literature reports per-window; our primary evaluation reports per-recording (the clinically meaningful metric). Comparing the two without care manufactures false wins.

What is transfer?

TUAB is a binary task (normal / abnormal). TUEV is a 6-class task (spike-and-slow-wave, generalised periodic discharge, lateralised periodic discharge, eye movement, artefact, background).

Transfer means: train the encoder on TUAB without labels, then freeze it and train a linear probe on TUEV. If the representations are genuinely general — not just memorised TUAB-specific patterns — they should also separate TUEV event types, even though the encoder never saw a TUEV label.

A model that transfers well is learning something about the *structure of EEG* rather than the specific abnormality label.

Why does the simplest encoder win?

A 0.4-million-parameter strided convolutional network (Conv1D) consistently outperforms larger transformer architectures (1.2M–3.65M parameters) in our experiments.

The intuition: transformers are powerful but data-hungry. They need millions of examples and long training to learn useful attention patterns. On 2 717 recordings trained for 20 epochs, there is simply not enough signal to fill their capacity. The small conv has less capacity but fits the data regime.

The published foundation models (LaBraM, BIOT) that do outperform our conv were pretrained on tens of thousands of hours of EEG data across multiple hospitals and devices — a scale we do not have access to.